Advanced Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Distance, midpoint, and slope combined**

Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB.

Midpoint of AB: $\left(\frac{-2+4}{2}, \frac{1+1}{2}\right) = (1, 1)$. Length of median from $C(1, 7)$ to $(1, 1)$: $\sqrt{(1-1)^2 + (7-1)^2} = \sqrt{0 + 36} = 6$.

2 **[Foundational] • Target: Equations of circles from given conditions**

Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.

Center (midpoint): $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$. Radius (half the diameter length): diameter $= \sqrt{(8-2)^2 + (11-3)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$, so radius $= 5$. Equation: $(x-5)^2 + (y-7)^2 = 25$.

3 **[Foundational] • Target: Distance from a point to a line**

Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

$d = \frac{|3(3) - 4(4) + 5|}{\sqrt{9 + 16}} = \frac{|9 - 16 + 5|}{5} = \frac{|-2|}{5} = \frac{2}{5}$.

4 **[Foundational] • Target: Locus problems**

Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.

The locus of points equidistant from two fixed points is the perpendicular bisector of the segment joining them. Since both points lie on the x-axis symmetric about the origin, the perpendicular bisector is the vertical line $x = 0$ (the y-axis).

5 **[Foundational] • Target: Rotations and general transformations**

A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \to (-y, x)$.

Applying the rule: $(4, 0) \to (-0, 4) = (0, 4)$.

6 **[Foundational] • Target: Dilations and scale factor**

A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.

Multiply each coordinate by 3: $(3, 6)$, $(9, 6)$, $(3, 18)$.

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7 [Foundational] • Target: Finding the equation of a parabola from three points

A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a, b, and c.

From $(0, 3)$: $c = 3$. From $(1, 6)$: $a + b + c = 6$, so $a + b = 3$. From $(-1, 2)$: $a - b + c = 2$, so $a - b = -1$. Adding $a + b = 3$ and $a - b = -1$: $2a = 2$, so $a = 1$, then $b = 2$. Equation: $y = x^2 + 2x + 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: A multi-part triangle-in-the-plane problem

This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula Area $= \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

(a) Area $= \frac{1}{2}|0(0 - 6) + 6(6 - 0) + 3(0 - 0)| = \frac{1}{2}|0 + 36 + 0| = 18$. (b) Using base $PQ = 6$ (along the x-axis) and height equal to R's y-coordinate, 6: Area $= \frac{1}{2}(6)(6) = 18$ ✓, confirming both methods agree.

9 [Intermediate] • Target: Finding the equation of a line given a point and a perpendicular condition

Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.

Midpoint: $\left(\frac{2+8}{2}, \frac{3+7}{2}\right) = (5, 5)$. Slope of the segment: $\frac{7-3}{8-2} = \frac{4}{6} = \frac{2}{3}$. Perpendicular slope: $-\frac{3}{2}$. Equation: $y - 5 = -\frac{3}{2}(x - 5)$, so $y = -\frac{3}{2}x + \frac{15}{2} + 5 = -\frac{3}{2}x + \frac{25}{2}$.

10 [Intermediate] • Target: Translating and reflecting equations of curves

The graph of $y = x^2$ is reflected over the x-axis, then shifted 3 units up. Write the equation of the resulting graph.

Reflecting over the x-axis: $y = -x^2$. Shifting up 3: $y = -x^2 + 3$.

11 [Intermediate] • Target: Finding vertices of a polygon given transformations

A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.

Add $(3, -2)$ to each vertex: $(3, -2)$, $(7, -2)$, $(7, 2)$, $(3, 2)$.

12 [Intermediate] • Target: Finding the intersection of a line and a parabola

Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

Set equal: $x^2 - 2x - 3 = x - 3$, so $x^2 - 3x = 0$, giving $x(x - 3) = 0$, so $x = 0$ or $x = 3$. Corresponding y-values: for $x = 0$, $y = -3$; for $x = 3$, $y = 0$. Intersection points: $(0, -3)$ and $(3, 0)$.

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13 **[Intermediate]** • **Target: A multi-part quadrilateral classification problem**

This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0,0)$, $X(6,0)$, $Y(8,4)$, $Z(2,4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY, and of WZ and XY. (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

(a) $WX = \sqrt{(6-0)^2+0^2} = 6$. $XY = \sqrt{(8-6)^2+4^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$. $YZ = \sqrt{(2-8)^2+0^2} = 6$. $ZW = \sqrt{(0-2)^2+(-4)^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$. (b) Slope of WX: $\frac{0-0}{6-0} = 0$. Slope of ZY: $\frac{4-4}{8-2} = 0$ (parallel, since equal slopes). Slope of WZ: $\frac{4-0}{2-0} = 2$. Slope of XY: $\frac{4-0}{8-6} = 2$ (also parallel, equal slopes). (c) Since both pairs of opposite sides are parallel ($WX \parallel ZY$ and $WZ \parallel XY$) and opposite sides are equal in length (.

14 **[Intermediate]** • **Target: Finding the equation of a tangent line to a circle**

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3,4)$, using the fact that the tangent is perpendicular to the radius at that point.

The radius to $(3,4)$ has slope $\frac{4-0}{3-0} = \frac{4}{3}$. The tangent line is perpendicular, so its slope is $-\frac{3}{4}$. Equation: $y - 4 = -\frac{3}{4}(x-3)$, so $y = -\frac{3}{4}x + \frac{9}{4} + 4 = -\frac{3}{4}x + \frac{25}{4}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Finding the vertex and axis of symmetry of a parabola from three points** • **Cross-Lesson: 7.01 Advanced Plane Geometry**

A parabola passes through $(-1,0)$, $(3,0)$, and $(1,-8)$. Find the axis of symmetry and the vertex.

Since the parabola has zeros at $x = -1$ and $x = 3$, the axis of symmetry lies midway between them: $x = \frac{-1+3}{2} = 1$. Since $(1,-8)$ lies exactly on the axis of symmetry $x = 1$, this point IS the vertex: $(1,-8)$.

16 **[Challenging]** • **Target: Distance, midpoint, and slope combined** • **Cross-Lesson: 8.01 Statistics, Counting & Probability**

Find the standard equation $Ax + By = C$ of the line perpendicular to $3x - 4y = 12$ that passes through the midpoint of the line segment joining $P(-2,7)$ and $Q(6,1)$.

(a) Midpoint M: $M = \left(\frac{-2+6}{2}, \frac{7+1}{2}\right) = \left(\frac{4}{2}, \frac{8}{2}\right) = (2,4)$. (b) Perpendicular slope: Rewrite the given line in slope-intercept form: $4y = 3x - 12 \Rightarrow y = \frac{3}{4}x - 3$. The slope is $m = \frac{3}{4}$. The perpendicular slope is the negative reciprocal: $m_\perp = -\frac{4}{3}$. (c) Line equation: Using point-slope form with point $(2,4)$.

17 **[Challenging]** • **Target: Equations of circles from given conditions** • **Cross-Lesson: 1.01 Advanced Number Theory**

Find the shortest perpendicular distance from the point $P(2,-3)$ to the line $3x - 4y + 7 = 0$.

Apply the point-to-line perpendicular distance formula $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2+B^2}}$ with $A = 3$, $B = -4$, $C = 7$, and $(x_0, y_0) = (2,-3)$: $d = \frac{|3(2)+(-4)(-3)+7|}{\sqrt{3^2+(-4)^2}} = \frac{|6+12+7|}{\sqrt{9+16}} = \frac{|25|}{\sqrt{25}} = \frac{25}{5} = \mathbf{5}$

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18 [Challenging] • Target: Distance from a point to a line • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the distance between points $A(1, 2, 3)$ and $B(4, 6, 3)$, and find the midpoint M of the line segment AB.

(a) Distance in 3D: $d = \sqrt{(4-1)^2 + (6-2)^2 + (3-3)^2} = \sqrt{3^2 + 4^2 + 0^2} = \sqrt{9 + 16 + 0} = \sqrt{25} = \mathbf{5}$ (b) Midpoint: $M = \left(\frac{1+4}{2}, \frac{2+6}{2}, \frac{3+3}{2}\right) = \left(\mathbf{\frac{5}{2}, 4, 3}\right)$

19 [Challenging] • Target: Locus problems • Cross-Lesson: 3.01 Advanced Functions

An ellipse has the equation $9x^2 + 25y^2 = 225$. Determine its semi-major axis a, semi-minor axis b, the coordinates of its foci, and its eccentricity e.

Step 1 (Standard Form): Divide both sides by 225: $\frac{9x^2}{225} + \frac{25y^2}{225} = 1 \Rightarrow \frac{x^2}{25} + \frac{y^2}{9} = 1$ Step 2 (Axes and Foci): Since the denominator of x^2 is larger, the major axis is horizontal: $a^2 = 25 \Rightarrow a = \mathbf{5}$ $b^2 = 9 \Rightarrow b = \mathbf{3}$ Focal distance: $c = \sqrt{a^2 - b^2} = \sqrt{25 - 9} = \sqrt{16} = \mathbf{4}$. Foci coordinates: $(\pm c, 0) = \mathbf{(\pm 4, 0)}$. Step 3 (Eccentricity): $e = \frac{c}{a} = \frac{4}{5} = \mathbf{0.8}$.

20 [Challenging] • Target: Rotations and general transformations • Cross-Lesson: 4.01 Advanced Trigonometry

A quadrilateral $ABCD$ is inscribed in a circle. If $\angle DAB = (2x + 15)^\circ$ and the opposite angle $\angle DCB = (3x - 10)^\circ$, determine the value of x and calculate the measure of $\angle DAB$.

(a) Supplementary property: Opposite angles of any cyclic quadrilateral sum to 180°: $(2x + 15) + (3x - 10) = 180 \Rightarrow 5x + 5 = 180 \Rightarrow 5x = 175 \Rightarrow \mathbf{x = 35}$. (b) Angle measure: Substitute $x = 35$ into $\angle DAB$: $\angle DAB = 2(35) + 15 = 70 + 15 = \mathbf{85}^\circ$. (Check: $\angle DCB = 3(35) - 10 = 105 - 10 = 95^\circ$, and $85^\circ + 95^\circ = 180^\circ$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distance, midpoint, and slope combined

Find the slope of the line through $(4, -5)$ and $(8, -2)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$ ✓

c. $-\frac{4}{3}$

d. $\frac{4}{3}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 + 5}{8 - 4} = \frac{3}{4}$.

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22 [Foundational] • Target: Equations of circles from given conditions

Find the y-intercept of the perpendicular bisector of the segment connecting $(-2, -1)$ and $(4, -6)$.

a. $-\frac{47}{10}$ ✓

b. $-\frac{8}{3}$

c. $-\frac{37}{10}$

d. $\frac{47}{10}$

Midpoint: $(1, -7/2)$. Slope of the segment: $-5/6$, so the perpendicular slope is $6/5$. Using $y + 7/2 = 6/5(x - 1)$, the y-intercept is $-\frac{47}{10}$.

23 [Foundational] • Target: Distance from a point to a line

Find the sum of the x-coordinates of the intersection points of $y = x^2 - 12$ and $y = -4x$.

a. -12

b. -4 ✓

c. -2

d. -6

Setting the equations equal: $x^2 + 4x - 12 = 0$. By Vieta's formulas, the sum of the roots is $-(4) = -4$.

24 [Foundational] • Target: Locus problems

In the standard (x, y) coordinate plane, the line $y = mx$ (where $m > 0$) is tangent to the circle $(x - 10)^2 + y^2 = 36$. What is the value of m?

a. $\frac{3}{5}$

b. $\frac{3}{4}$ ✓

c. $\frac{4}{5}$

d. $\frac{4}{3}$

We are given the circle $(x - 10)^2 + y^2 = 36$, which has its center at $C(10, 0)$ and radius $r = \sqrt{36} = 6$. **Method 1: Geometric Approach (Right Triangle)** Let $O(0, 0)$ be the origin and let T be the point of tangency in the first quadrant. A radius drawn to the point of tangency is perpendicular to the tangent line, so ΔOTC is a right triangle with right angle at T. - Hypotenuse: $OC = 10$ - Leg (radius): $CT = 6$ By the Pythagorean theorem.

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25 [Foundational] • Target: Rotations and general transformations

In the standard (x, y) coordinate plane, the vertices of ΔOAB are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides ΔOAB into two regions of equal area. What is the value of k?

a. $6 - 2\sqrt{3}$ ✓

b. $\sqrt{6}$

c. $\frac{8}{3}$

d. 3

1. **Find the total area of ΔOAB:** Taking the side along the x-axis as the base, the length of base OA is $6 - 0 = 6$. The height is the y-coordinate of vertex $B(2, 4)$, which is 4. $\text{Area}(\Delta OAB) = \frac{1}{2} \times 6 \times 4 = 12$ To divide the triangle into two regions of equal area, each region must have an area of $\frac{12}{2} = 6$. 2. **Determine the domain of k:** - If 02, the line $x = k$ lies to the right of vertex $B(2, 4)$. The region to the right of $x = k$ is a t.

26 [Foundational] • Target: Dilations and scale factor

In the standard (x, y) coordinate plane, line L is perpendicular to the line $3x + 4y = 15$ and passes through the point $(3, 1)$. Line L intersects the line $2x + y = 17$ at the point (p, q). What is the value of $p + q$?

a. 5

b. 6

c. 8

d. **11** ✓

Step 1: Find the slope of the given line $3x + 4y = 15$. Rewriting in slope-intercept form gives: $4y = -3x + 15 \Rightarrow y = -\frac{3}{4}x + \frac{15}{4}$ Thus, the slope of this line is $m_1 = -\frac{3}{4}$. Step 2: Determine the slope of line L. Since line L is perpendicular to the given line, its slope m_L is the negative reciprocal of m_1: $m_L = -\frac{1}{-\frac{3}{4}} = \frac{4}{3}$ Step 3: Write the equation of line L. Line L passes through $(3, 1)$ with slope $\frac{4}{3}$.

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27 [Foundational] • Target: Finding the equation of a parabola from three points

In the figure below, ΔABC is a right triangle with hypotenuse $\overline{AB}$, and altitude $\overline{CD}$ of length h is drawn to $\overline{AB}$. The altitude divides the hypotenuse into two segments of lengths $AD = 9$ and $DB = 16$. A circle with center O_1 is inscribed in ΔACD, and a circle with center O_2 is inscribed in ΔBCD. What is the distance between the centers O_1 and O_2?

a. 5

b. $5\sqrt{2}$ ✓

c. 7

d. $7\sqrt{2}$

We solve the problem in five systematic steps: 1. **Determine altitude CD**: By the Geometric Mean Theorem for right triangles, the altitude to the hypotenuse satisfies:
$CD^2 = AD \cdot DB = 9 \cdot 16 = 144 \Rightarrow CD = 12.$ 2. **Find the side lengths of ΔACD and ΔBCD**: - In right triangle ΔACD, the legs are $AD = 9$ and $CD = 12$. By the Pythagorean theorem:
$AC = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$ - In right triangle ΔBCD, the legs are $DB = 16$ an.

28 [Foundational] • Target: A multi-part triangle-in-the-plane problem

In the standard (x, y) coordinate plane, what are the center and the radius of the circle with equation $(x - 4)^2 + (y + 6)^2 = 49$?

a. **Center** $(4, -6)$**, radius** 7 ✓

b. Center $(-4, 6)$, radius 7

c. Center $(4, -6)$, radius 49

d. Center $(-4, 6)$, radius 49

The standard form of the equation of a circle in the (x, y) coordinate plane is: $(x - h)^2 + (y - k)^2 = r^2$ where (h, k) is the center of the circle and r is the radius (with $r > 0$). Comparing the given equation $(x - 4)^2 + (y + 6)^2 = 49$ to the standard form: 1. For the x-coordinate of the center:
$x - h = x - 4 \Rightarrow h = 4$. 2. For the y-coordinate of the center: $y - k = y + 6 = y - (-6) \Rightarrow k = -6$.
Therefore, the center is $(4, -6)$. 3. For the radius.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Finding the equation of a line given a point and a perpendicular condition

In the standard (x, y) coordinate plane, a circle has its center at $(-3, 5)$ and passes through the point $(1, 8)$. What is the radius of this circle?

a. 5 ✓

b. 7

c. $\sqrt{13}$

d. 25

The radius r of a circle is the distance from its center to any point on the circle. Using the distance formula between the center $(-3, 5)$ and the point on the circle $(1, 8)$: $r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $r = \sqrt{(1-(-3))^2 + (8-5)^2} = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$ Thus, the radius of the circle is 5. Common misconceptions: - Choice B (7) mistakenly adds the coordinate differences directly: $4 + 3 = 7$. - Choice C $(\sqrt{13})$ makes a sign error in subtracting the.

30 [Intermediate] • Target: Translating and reflecting equations of curves

In the standard (x, y) coordinate plane, what is the length of the major axis of the ellipse defined by the equation $\frac{(x-4)^2}{49} + \frac{(y+2)^2}{16} = 1$?

a. 7

b. 8

c. **14** ✓

d. 49

The standard form of a horizontal ellipse is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, where $a > b > 0$. 1. Identify the values of a^2 and b^2 from the given equation: $a^2 = 49 \quad \text{and} \quad b^2 = 16$ 2. Solve for the semi-major axis length a: $a = \sqrt{49} = 7$ 3. The total length of the major axis is $2a$: $2a = 2 \times 7 = 14$ Common misconceptions: - Choosing 7 confuses the semi-major axis length a with the full major axis length $2a$. - Choosing 8 calculates the length of the.

31 [Intermediate] • Target: Finding vertices of a polygon given transformations

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. **25** ✓

b. 50

c. 100

d. 125

The equation of the circle is $x^2 + y^2 = 25$. Let the point of tangency be $P(x_0, y_0)$. The equation of the tangent line to the circle at $P(x_0, y_0)$ is given by $xx_0 + yy_0 = 25$. To find the x-intercept A, we set $y = 0$: $xx_0 = 25 \Rightarrow x = \frac{25}{x_0}$. So, $A = (\frac{25}{x_0}, 0)$. To find the y-intercept B, we set $x = 0$: $yy_0 = 25 \Rightarrow y = \frac{25}{y_0}$. So, $B = (0, \frac{25}{y_0})$. The area of triangle OAB is.

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32 [Intermediate] • Target: Finding the intersection of a line and a parabola

In the xy-plane, let $P(t, t^2)$ with $t > 0$ be a point on the parabola $y = x^2$. The normal line to the parabola at P intersects the parabola again at a second point Q. Let T be the intersection point of the tangent lines to the parabola at P and Q. As $t \to 0^+$, what is the limiting value of the y-coordinate of T?

a. $-\frac{1}{2}$ ✓

b. $-\frac{1}{4}$

c. -1

d. 0

We synthesize differential calculus, coordinate geometry, and limits to determine the behavior of T as $t \to 0^+$. 1. **Equation of the Normal Line at P:** Since $y = x^2$, the derivative is $y' = 2x$. At point $P(t, t^2)$, the slope of the tangent line is $2t$. Therefore, the slope of the normal line is the negative reciprocal: $m_{\text{normal}} = -\frac{1}{2t}$ The equation of the normal line at P is: $y - t^2 = -\frac{1}{2t}(x - t)$ 2. **Finding the Second Intersection Point Q:** To find wher.

33 [Intermediate] • Target: A multi-part quadrilateral classification problem

A circle has center $(4, -2)$ and passes through $(13, 10)$. Find the slope of the tangent line to the circle at $(13, 10)$.

a. 5/4

b. 1/4

c. **-3/4** ✓

d. -11/4

Radius slope $= \frac{10+2}{13-4} = 4/3$. Tangent $\perp$ radius, so tangent slope $= -\frac{1}{4/3} = -3/4$.

34 [Intermediate] • Target: Finding the equation of a tangent line to a circle

In ΔABC, the side lengths are $AB = 9$, $AC = 8$, and $BC = 7$. A line segment $\overline{DE}$, with D lying on $\overline{AB}$ and E lying on $\overline{AC}$, divides ΔABC into two regions that have equal areas and equal perimeters. What is the length of $\overline{DE}$?

a. $2\sqrt{6}$ ✓

b. $\frac{7\sqrt{2}}{2}$

c. $4\sqrt{3}$

d. $6\sqrt{2}$

The line segment $\overline{DE}$ divides ΔADE into two regions: ΔADE and quadrilateral $BDEC$. 1. **Equal Perimeters:** The perimeter of ΔADE is $AD + AE + DE$. The perimeter of quadrilateral $BDEC$ is $DB + BC + CE + DE = (AB - AD) + BC + (AC - AE) + DE$. Equating the two perimeters and subtracting DE from both sides gives: $AD + AE = (AB + BC + AC) - (AD + AE)$
$2(AD + AE) = 9 + 7 + 8 = 24 \Rightarrow AD + AE = 12$ 2. **Equal Areas:** Using the sine formula for area.

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35 [Intermediate] • Target: Finding the vertex and axis of symmetry of a parabola from three points

Each interior angle of a regular polygon measures 144°. How many sides does the polygon have?

a. 8

b. 9

c. **10** ✓

d. 12

To find the number of sides n of a regular polygon from its interior angle measure, we can use the relationship between the interior angle and the exterior angle. Each exterior angle of a regular polygon is supplementary to its corresponding interior angle: $\text{Exterior Angle} = 180^\circ - 144^\circ = 36^\circ$ The sum of the exterior angles of any convex polygon is always 360°. Since all n exterior angles in a regular polygon are equal: $n = \frac{360^\circ}{36^\circ} = 10$ Alternatively, using t.

36 [Intermediate] • Target: Distance, midpoint, and slope combined

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.

a. **25** ✓

b. 30

c. 35

d. 40

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$. Since point $P(a, 4)$ lies on the circle, we have $a^2 + 4^2 = r^2$, so $a^2 + 16 = r^2$. The radius from the origin to $P(a, 4)$ is perpendicular to the tangent line at P. The slope of the radius is $m_{radius} = \frac{4-0}{a-0} = \frac{4}{a}$. The given tangent line is $3x + 4y = k$, which can be rewritten in slope-intercept form as $y = -\frac{3}{4}x + \frac{k}{4}$. The slope of the tangent line is $m_{tangent} = -\frac{3}{4}$. Since the rad.

37 [Intermediate] • Target: Equations of circles from given conditions

In right triangle ABC, altitude $\overline{BD}$ is drawn to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of altitude $\overline{BD}$?

a. **6** ✓

b. 6.5

c. 13

d. 36

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments into which it divides the hypotenuse: $BD^2 = AD \cdot DC$ Substitute the given lengths $AD = 4$ and $DC = 9$: $BD^2 = 4 \cdot 9 = 36$ Taking the positive square root: $BD = \sqrt{36} = 6$ Common Misconceptions: - Choice B (6.5) confuses the geometric mean with the arithmetic mean: $\frac{4+9}{2} = 6.5$. - Choice C (13) incorrectly adds the two segments together: $4 + 9 = 13$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Distance from a point to a line** • **Cross-Lesson: 2.01 Advanced Algebra & Systems**

Find the radius of the circle $x^2 + y^2 + 4x + 12y + 15 = 0$.

a. **5** ✓

b. 25

c. 6

d. 4

Complete the square: $(x + 2)^2 + (y + 6)^2 = 25$, so $r = \sqrt{25} = 5$.

39 **[Challenging]** • **Target: Locus problems** • **Cross-Lesson: 3.01 Advanced Functions**

Find the distance between $(-2,\ -1)$ and $(-6,\ -4)$.

a. **5** ✓

b. 6

c. 4

d. 7

$d = \sqrt{(-6 + 2)^2 + (-4 + 1)^2} = \sqrt{4^2 + 3^2} = 5$.

40 **[Challenging]** • **Target: Rotations and general transformations** • **Cross-Lesson: 4.01 Advanced Trigonometry**

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. **4** ✓

c. 5

d. 6

The given parabola is $y = x^2 - 6x + 13$. We can rewrite this equation in vertex form by completing the square: $y = (x^2 - 6x + 9) + 13 - 9$, which simplifies to $y = (x - 3)^2 + 4$. This is the equation of a parabola with vertex $(h, k) = (3, 4)$, opening upwards. The standard form of a parabola opening upwards is $(x - h)^2 = 4p(y - k)$, where (h, k) is the vertex and p is the distance from the vertex to the focus and from the vertex to the directrix. Comparing our equation $(x - 3)^2 = 1(y - 4)$ with the s.

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41 [Challenging] • Target: Dilations and scale factor • Cross-Lesson: 5.01 Pre-Calculus Concepts

A circle $\mathcal{C}$ with center $C(0, k)$ on the positive y-axis passes through the focus F of the parabola $x^2 = 4y$. If $\mathcal{C}$ is tangent to the parabola at two distinct points, P and Q, what is the area of quadrilateral $PFQC$?

a. $8\sqrt{3}$ ✓

b. $4\sqrt{3}$

c. $16\sqrt{3}$

d. $6\sqrt{3}$

1. **Identify the focus and circle equation:** The standard form of the parabola is $x^2 = 4py$, so with $4p = 4$, we have $p = 1$. Thus, the focus is $F(0, 1)$. The circle is centered at $C(0, k)$ with $k > 1$ and passes through $F(0, 1)$, so its radius is $R = k - 1$. The equation of $\mathcal{C}$ is: $x^2 + (y - k)^2 = (k - 1)^2$ 2. **Find the tangency condition:** Substitute $x^2 = 4y$ from the parabola into the circle's equation: $4y + y^2 - 2ky + k^2 = k^2 - 2k + 1$ $y^2 - 2(k - 2)y + (2k - 1) = 0$ For.

42 [Challenging] • Target: Finding the equation of a parabola from three points • Cross-Lesson: 7.01 Advanced Plane Geometry

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

43 [Challenging] • Target: A multi-part triangle-in-the-plane problem • Cross-Lesson: 8.01 Statistics, Counting & Probability

A circle has center $(6, 6)$ and radius 2. Find the shortest distance from the point $(16, 30)$ to the circle.

a. 6

b. 38

c. **24** ✓

d. 29

Distance from point to center $= \sqrt{10^2 + 24^2} = 26$. Since $26 > 2$, the point is outside; shortest distance to the circle $= 26 - 2 = 24$.

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44 [Challenging] • Target: Finding the equation of a line given a point and a perpendicular condition • Cross-Lesson: 1.01 Advanced Number Theory

Line ℓ_1 has slope -2. What is the slope of a line perpendicular to ℓ_1?

a. 1/2 ✓

b. 5/2

c. 1/4

d. 1/6

Perpendicular slopes are negative reciprocals: $m = -\frac{1}{-2} = 1/2$.

45 [Challenging] • Target: Translating and reflecting equations of curves • Cross-Lesson: 2.01 Advanced Algebra & Systems

The line $y = -12$ intersects the circle $x^2 + y^2 = 169$ at two points. What is the distance between the two intersection points?

a. 10 ✓

b. 13

c. 16

d. 5

$x^2 = 169 - (-12)^2 = 25 \Rightarrow x = \pm 5$. Distance $= 2(5) = 10$.

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